

Experiment	GAT	Features	BiCA	MAE (m)	RMSE (m)	ADE (m)	FDE (m)	RMSE Improve	ADE Improve	FDE Improve
Baseline GRU (No Enhancements)	×	×	×	0.1823	0.2249	0.1823	0.4135	+0.0%	+0.0%	+0.0%
Features + BiCA (No GAT)	×	✓	✓	0.1190	0.1442	0.1190	0.2550	+35.9%	+34.7%	+38.3%
GAT + BiCA (No Features)	✓	×	✓	0.1785	0.2200	0.1785	0.4033	+2.2%	+2.1%	+2.5%
GAT + Features (No BiCA)	✓	✓	×	0.1154	0.1397	0.1154	0.2476	+37.9%	+36.7%	+40.1%
Full Model (DG32-BCAT)	✓	✓	✓	0.0990	0.1192	0.0990	0.2064	+47.0%	+45.7%	+50.1%

Ablation Study: Comprehensive Metrics Comparison